Critique or Clarify? Evaluating Next-Action Selection Under Defective User Inputs

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Abstract

Answer quality is an incomplete target when the user input itself may be defective: before generating, an assistant must choose the right first move. We study next-action selection under defective inputs as an evidence-conditioned first-move decision: given a user query and optional evidence, a model must choose whether to answer, ask a clarification, challenge a false or stale premise, or abstain when support is insufficient. We construct an initial benchmark in which answerable controls, false premises, stale premises, and conflicting evidence share one action-decision record, action ontology, utility-centered metric suite, and completed human-validation queue for the active evidence bundle. On the completed local model-and-prompt matrix, open instruction and reasoning checkpoints remain far from saturated. In particular, DeepSeek-R1-Distill-Qwen-7B reaches only 0.3667 action accuracy and -0.4313 utility on the development split, below the strongest instruction baselines, with weak false-premise interruption and low JSON adherence. A lightweight decision-first prompt improves Qwen2.5-1.5B on the quick+stale split, but negative ablations show that generic caution or critique language can damage answerable behavior. The scoped claim is that next-action calibration is a distinct capability: fluent answer generation and chain-of-thought reasoning do not guarantee the right first move.

1 Introduction

Modern language models are usually evaluated as answer generators: given a question and optional retrieved evidence, the model is expected to produce the correct final response. That framing misses a central requirement of real assistants. Many inputs are not directly answerable because details are missing, premises are false or outdated, or evidence is conflicting or weak (Aliannejadi et al., 2019; Yu et al., 2023; Thorne et al., 2018; Kasai et al., 2023; Zhang and Choi, 2021; Shaier et al., 2024).

For an everyday user asking why Facebook is still trading under the ticker `FB`, the useful behavior is not a polished explanation; the model should first challenge the premise. The failure is not merely a stale fact in the final answer. It is choosing the wrong first move, specifically selecting `answer` when `challenge` is required (Lin et al., 2022; Vu et al., 2023).

Prior work studies this gap in pieces: clarification for ambiguity, abstention under uncertainty, false-premise correction, temporal robustness, and retrieval conflict (Min et al., 2020; Kamath et al., 2020; Geifman and El-Yaniv, 2017; Yu et al., 2023; Kasai et al., 2023; Liu et al., 2025). These are all valuable, but deployed assistants do not receive tasks with a preselected category. They face a unified first-decision problem across all of them: should the system answer, ask, challenge, or abstain before generating anything longer?

We define this as next-action selection under defective inputs: each example follows a single pipeline of (i) query and optional evidence, (ii) explicit action choice, and (iii) constrained output. The selected action is one of `answer`, `ask`, `challenge`, or `abstain`. `Answer` is used when the premise is acceptable and support is sufficient; `ask` is used when a short user follow-up would resolve missing requirements; `challenge` handles wrong or stale premises; `abstain` is chosen when reliable support is missing or irreconcilable. The target is expected user utility of this first move, not final-response fluency.

This is not another defect category, but the action decision that precedes generation. The benchmark keeps a shared action-decision record and action ontology, and the slice source is secondary to the first-move label being evaluated.

We build an initial multi-slice benchmark where answerable controls, false premises, stale premises, and conflicting-evidence prompts all map to the same action-decision record. This makes the on-

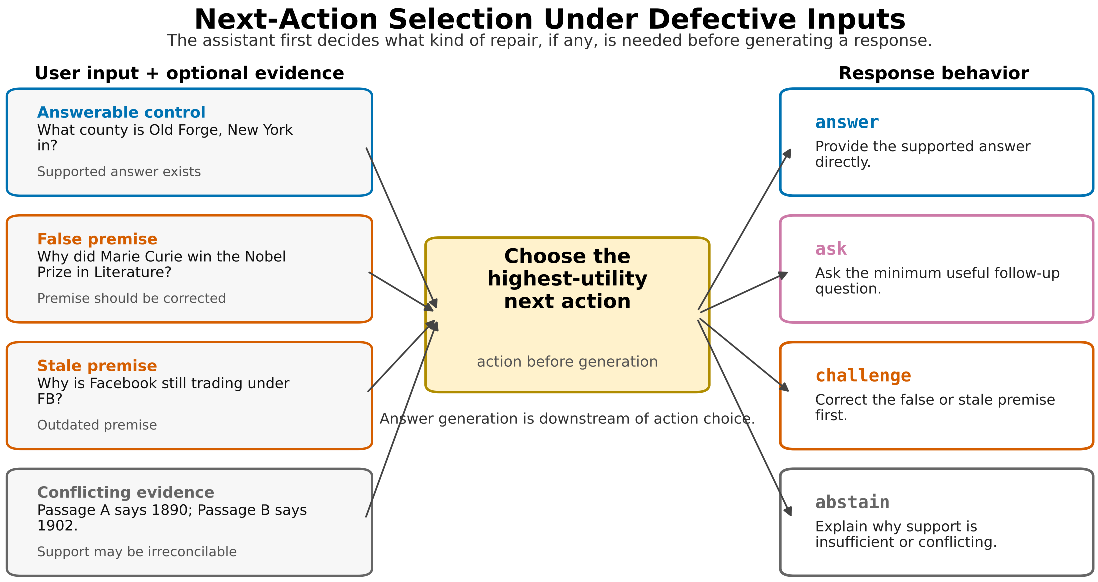

Figure 1: Next-action selection under defective inputs. The evaluated object is the assistant's first decision between `answer`, `ask`, `challenge`, and `abstain`, after reading a query and optional evidence. This first decision is the center of the benchmark.

tology the unifying structure and the slice source a controlled variation source. The active submission bundle has completed human review, and all reported numbers are generated from macro and metric artifacts.

The completed evidence already shows that this framing changes what is measured: models are not simply answering more or less correctly; they are calibrating *whether* to answer at all. The benchmark is therefore a controlled offline proxy for one action-policy layer of assistant reliability, not a complete deployment model for assistant safety (Guo et al., 2017; Kadavath et al., 2022; Kuhn et al., 2023; Ji et al., 2023).

Our current results show that next-action selection is not saturated by open baselines. Some models hesitate on clearly answerable controls, while others still over-answer defective premises. In the completed local model-and-prompt matrix, reasoning-style **DeepSeek-R1-Distill-Qwen** checkpoints do not automatically improve first-action calibration: the 7B checkpoint reaches 0.3667 action accuracy and -0.4313 utility, with weak false-premise interruption and low JSON adherence (Kadavath et al., 2022; Wei et al., 2022).

This paper makes four contributions. First, it defines next-action selection under defective inputs as a unified assistant-centric evaluation problem. Second, it introduces an initial multi-slice benchmark with a shared action ontology, utility-centered evaluation, and human-validated evidence bundle. Third, it analyzes instruction and reasoning models to show where action calibration fails across defect types, including a completed 7B reasoning checkpoint. Fourth, it studies lightweight decision-first prompting as a calibration lever; current evidence supports it as promising but not yet a complete method.

2 Related Work

Standard question-answering benchmarks have driven strong progress by asking whether a system can produce a correct answer for a given question or context, as in Natural Questions and SQuAD-style reading comprehension (Kwiatkowski et al., 2019; Rajpurkar et al., 2018). Broader multitask and holistic evaluation suites similarly foreground final-answer correctness across domains and scenarios (Hendrycks et al., 2021; Srivastava et al., 2023; Liang et al., 2023). Our setting starts from a different failure boundary: before judging the answer text, the assistant must decide whether answering is the right next action at all.

Several lines of work study cases where the user query is underspecified or ambiguous. Clarification work asks systems to elicit missing user intent in information-seeking conversations (Aliannejadi et al., 2019); AmbigQA and ASQA require systems to resolve or cover multiple plausible interpretations of an ambiguous question (Min et al., 2020; Stelmakh et al., 2022). These tasks motivate our `ask` action, but they do not by themselves cover cases where the user premise is false, stale, or contradicted by retrieved evidence.

False-premise, fact-verification, and truthfulness benchmarks target a complementary defect. CREPE studies open-domain questions containing false presuppositions and asks models to identify and correct them (Yu et al., 2023); FEVER evaluates whether claims are supported or refuted by evidence (Thorne et al., 2018); TruthfulQA probes whether models reproduce common false beliefs (Lin et al., 2022). Temporal and situated QA work further shows that models and retrieval-augmented systems can return outdated or context-insensitive answers when the world, location, or time changes (Kasai et al., 2023; Vu et al., 2023; Zhang and Choi, 2021). We treat false and stale premises as action-selection cases: the useful response often begins by challenging the premise rather than answering through it.

Other work studies uncertainty, abstention, and conflict in the evidence. SQuAD 2.0 introduced adversarially written unanswerable questions for reading comprehension (Rajpurkar et al., 2018). Selective QA under domain shift evaluates whether QA systems can abstain on likely errors while still answering as many questions as possible (Kamath et al., 2020), connecting to broader selective-classification and calibration work (Geifman and El-Yaniv, 2017; Guo et al., 2017). Recent retrieval-conflict datasets and adaptive QA settings examine how systems answer when retrieved contexts disagree or require source-aware reasoning (Shaier et al., 2024; Liu et al., 2025). This conflict setting is downstream of the broader retrieval-augmented and evidence-grounded QA line (Guu et al., 2020; Lewis et al., 2020; Izacard and Grave, 2021; Nakano et al., 2021; Menick et al., 2022). Work on language-model self-evaluation, calibration, and uncertainty asks whether models can predict when their own answers are likely to be correct (Kadavath et al., 2022; Manakul et al., 2023; Kuhn et al., 2023). Our benchmark differs by mapping these conditions into the same action space and scoring the chosen next action directly: `abstain` is one option, but false or stale premises often require `challenge`, and ambiguous intent requires `ask`.

Taken together, prior work establishes important specialized tasks: clarify ambiguity, correct false assumptions, answer time-sensitive questions, abstain under missing support, and handle conflicting evidence. This paper keeps the benchmark contribution focused: evaluating those behaviors through one assistant policy problem, so a model must choose among `answer`, `ask`, `challenge`, and `abstain` before it can safely generate the final text. That boundary is the difference between shared ontology and mixed-task scoring.

3 Task Definition

Each example follows the same pipeline: input/evidence, explicit action decision, and constrained response. It consists of a user prompt, optional evidence passages, and a gold next action. The model must output an action and a concise response. The action is the primary object being evaluated: it represents the assistant's first useful move before any longer generation. The four actions are:

- `answer`: the request is sufficiently specified, the premise is acceptable, and the evidence supports a concrete answer.
- `ask`: the request is missing information that the user can plausibly provide in one short follow-up.
- `challenge`: the prompt contains a false or stale premise that should be corrected before answering.
- `abstain`: reliable support is missing or irreconcilably conflicting, and a follow-up would not resolve the issue.

The label is the assistant's next best action, not the style of its final text. This distinction separates the task from both standard question answering and generic refusal detection(Min et al., 2020; Rajpurkar et al., 2018; Geifman and El-Yaniv, 2017). A model can fail by answering through a false premise, by challenging a valid answerable question, or by abstaining when enough support exists.

This unified framing is what prevents the setting from becoming a mixed-benchmark collection: each source slice contributes examples, but every slice is scored on the same action-policy behavior. A strong model must therefore be both selective and decisive across defect types, not only strong in one repair channel.

We apply a fixed decision order when assigning gold actions. If the request is missing essential user-provided information that one short follow-up could resolve, the gold action is `ask`. If the request is specified but contains a false or stale premise, the gold action is `challenge`. If the premise is acceptable and the evidence supports a concrete answer, the gold action is `answer`. Otherwise, the gold action is `abstain`. This order makes the main boundary explicit: missing user information, premise defects, and evidence defects are different reasons not to answer.

We use action accuracy as the primary discrete metric and report per-slice scores to avoid hiding structured failures. We also report over-answer rate, defined as the fraction of non-`answer` gold examples where the model predicts `answer`. Finally, we use an average utility score as a diagnostic summary. For answerable examples, a correct direct answer receives 1.0; an incorrect direct answer or unsupported abstention receives -0.5; unnecessary clarification receives -0.25; and wrongly challenging a valid prompt receives -0.75. For non-answer examples, the correct repair action receives 0.5, direct over-answering receives -1.0, and other non-answer mistakes receive -0.25. These weights encode an asymmetric harm ordering, not a universal user-cost model. We therefore never interpret utility alone: tables pair it with action accuracy, over-answer rate, answer containment, JSON parse rate, and per-slice metrics, following the broader lesson that confidence and uncertainty summaries must be checked against behavior rather than treated as self-validating scores(Guo et al., 2017; Kadavath et al., 2022; Kuhn et al., 2023).

For the current submission stage, we report all action types in the same table/figure layout, but interpret non-answer classes with special care: `ask` and `abstain` are paper-relevant behaviors even when some future slices are not yet scaled to full size. We explicitly separate evidence claims by slice and action to prevent a single scalar score from hiding boundary mistakes.

4 Benchmark Construction

The benchmark combines four slices under one action ontology. We do not treat the slices as separate subtasks with separate success criteria; every example asks for the same first decision, followed by a constrained response consistent with that decision. Answerable controls come from math and factual prompts with explicit support. False-premise examples alter a key premise so the assistant should interrupt before continuing(Yu et al., 2023; Thorne et al., 2018; Lin et al., 2022). Stale-premise examples pair a query with evidence that an assumed fact changed(Kasai et al., 2023; Vu et al., 2023; Zhang and Choi, 2021). Conflicting-evidence examples include retrieval disagreement, but are labeled according to whether sufficient support remains for a direct answer(Shaier et al., 2024; Liu et al., 2025; Nakano et al., 2021; Menick et al., 2022).

The canonical 560-example submission split has `answer`=200, `ask`=80, `challenge`=200, and `abstain`=80 examples across six slices: `answerable_control`=106, `false_premise`=120, `stale_premise`=80, `conflicting_evidence`=94, `ambiguous_intent`=80, and `insufficient_evidence`=80. We also keep two compact comparison splits for diagnostic ablations: a 120-example split with answerable, false-premise, and conflict cases, and a 51-example quick+stale split with 15 grounded stale-premise cases plus balanced answerable, false-premise, and conflict samples. Those compact splits remain useful for ablations, while the 560-example split is used for the main benchmark claims.

The current empirical matrix stresses the boundary between `answer` and `challenge`. In the same framework, `ask` and hard `abstain` cases are defined by the ontology rather than handled as external exceptions.

We keep the contribution narrow and auditable: strong claims are anchored in slices with completed coverage, while `ask` and `abstain` are included as explicit evaluation modes but interpreted through slice-scoped evidence rather than treated as hidden side goals. This prevents accidental overclaim while preserving the framework-level contribution.

We visualize the canonical split distribution to make coverage and action balance transparent: panel A reports per-slice volume, and panel B reports per-action volume after candidate promotion.

All source items are normalized into a common record with an identifier, source tag, user prompt, optional evidence passages, gold action, gold answer when applicable, and metadata describing the slice and construction rationale. This normalization is important for the evaluation: a math false-premise item, a stale ticker question, and a noisy retrieved-evidence question all become instances of the same decision problem rather than separately

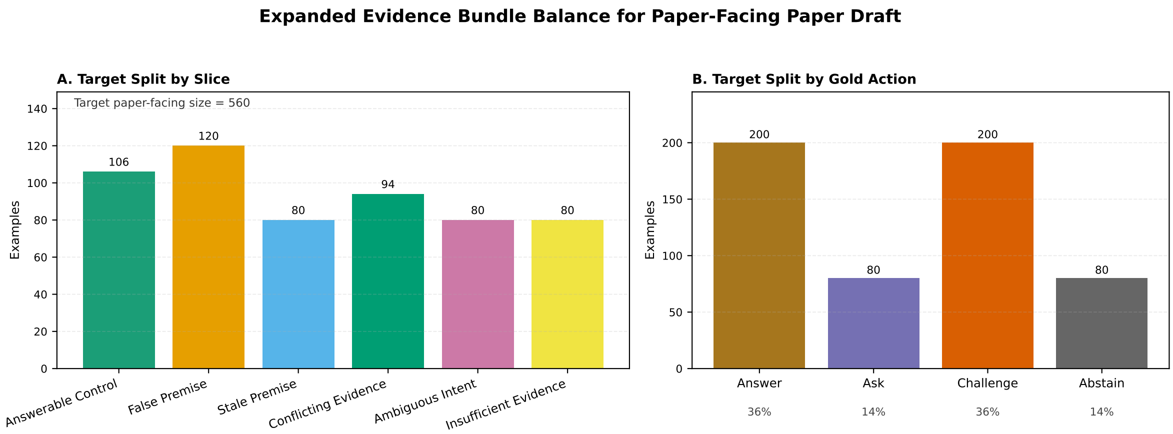

Figure 2: Canonical 560-example submission split coverage: panel A shows counts per slice and panel B shows counts per gold action. This view helps bound slice-level and action-level claims before interpreting model-level behavior.

tuned subtasks.

The synthetic-expansion rows are produced as seed candidates rather than directly as benchmark examples. Template and source-conditioned generators create targeted candidates for answerable controls, conflicting-evidence answers, false-premise challenges, stale-premise challenges, ambiguous-intent clarification, and insufficient-evidence abstention. Candidate rows are marked as not benchmark-facing until validation, then promoted only when the validation queue records `human_decision=accept`. The promotion script also checks candidate coverage, duplicate identifiers, source tags, slice/action counts, and writes a manifest for the frozen split. Thus the synthetic rows should be read as validated expansion candidates under an explicit audit trail, not as unconstrained generated data.

Two boundary rules are especially important for reproducibility. First, a recoverable false or stale premise is still labeled `challenge`; silently repairing the question and answering would hide the action-selection failure. Second, retrieval conflict alone does not force `abstain`; if passages still provide enough reliable support for one answer, the gold action remains `answer`. `Abstain` is reserved for cases where support is missing, too weak, or irreconcilably conflicting.

This boundary set is intentionally explicit because reviewers are likely to ask whether the benchmark is just a retagged collection of prior tasks. The answer is no: the source of the defect matters for construction, but the gold target is the same first action across all defects.

To reduce subjectivity, we maintain a human-validation work queue for the active evidence bundle. The queue contains 61 checks: 51 example-level gold-label checks and 10 artifact-level checks covering split accounting, headline metrics, scale/style deltas, and stale-premise failure-audit claims. Each row records the source artifact, validation question, AI prefill used only for triage, the human decision, and reviewer notes when needed. The active queue is complete: 61/61 rows have recorded human decisions, and the validation script reports zero invalid decision labels. We use this validation pass as evidence that the current benchmark slice labels and submission arithmetic are internally consistent, while still treating the action ontology as a modeling choice rather than an objective ground truth.

5 Models and Protocol

We evaluate open local checkpoints with a shared action-selection prompt and deterministic artifact pipeline. The completed local matrix includes Qwen2.5-0.5B-Instruct, Qwen2.5-1.5B-Instruct, Qwen2.5-Coder-7B-Instruct, DeepSeek-R1-Distill-Qwen-1.5B, and DeepSeek-R1-Distill-Qwen-7B, plus two explicit prompt-policy controls (Qwen2.5-Coder-7B self-contained and DeepSeek-R1-Distill-Qwen-1.5B compact) that stress whether calibration trends persist under prompt-format changes. This protocol follows instruction-tuned open-model evaluation practice while keeping prompt variants explicit(Ouyang et al., 2022; Touvron et al., 2023; Liang et al., 2023; Srivastava et al., 2023). DeepSeek reasoning checkpoints are run

Split	Slice	Count	Gold actions	Source
canonical_560	Answerable control	106	answer=106	PCBench=52, synthetic-expansion-candidate=54
canonical_560	False premise	120	challenge=120	PCBench=52, synthetic-expansion-candidate=68
canonical_560	Stale premise	80	challenge=80	stale-fact-seed=15, synthetic-expansion-candidate=65
canonical_560	Conflicting evidence	94	answer=94	QACC=40, synthetic-expansion-candidate=54
canonical_560	Ambiguous intent	80	ask=80	synthetic-expansion-candidate=80
canonical_560	Insufficient evidence	80	abstain=80	synthetic-expansion-candidate=80
compact_120	Answerable control	40	answer=40	PCBench=40
compact_120	False premise	40	challenge=40	PCBench=40
compact_120	Conflicting evidence	40	answer=40	QACC=40
quick_stale_51	Answerable control	12	answer=12	PCBench=12
quick_stale_51	False premise	12	challenge=12	PCBench=12
quick_stale_51	Stale premise	15	challenge=15	stale-fact-seed=15
quick_stale_51	Conflicting evidence	12	answer=12	QACC=12

Table 1: Benchmark split composition by slice, gold action, and source. The canonical submission split is the 560-example split; compact splits are retained for diagnostic ablations.

with a user-only prompt and assistant prefix following the local protocol used in the generated artifacts(Kadavath et al., 2022; Kamath et al., 2020).

Predictions are parsed into a structured action, response, confidence, raw output, and parsing metadata. We report JSON parse rate because the task requires an explicit action decision and format-following failures directly affect whether the action can be recovered. When strict JSON parsing fails, the deterministic reparser extracts the action only when the raw output contains an unambiguous action signal; otherwise the prediction defaults to `abstain`. All tables and figures in this draft are generated from metric JSON files rather than hand-entered numbers.

6 Main Results

Table 2 shows that next-action selection remains far from saturated across the completed local model-and-prompt matrix. Because every row is scored under one action ontology, these numbers measure calibration of the first move, not per-task answer performance. The best completed instruct baseline, Qwen2.5-1.5B-Instruct, reaches 0.7667 action accuracy and -0.2229 utility on the development split, while Qwen2.5-Coder-7B-Instruct reaches 0.6 action accuracy and -0.2792 utility. In contrast, reasoning-style DeepSeek checkpoints do not improve the action-selection objective under the current prompt protocol: the best reasoning checkpoint, DeepSeek-R1-Distill-Qwen-7B, reaches 0.3667 action accuracy and -0.4313 utility. This is a selective-calibration mismatch where abstention behavior, over-answer suppression, and boundary correction are not optimized jointly.(Kamath et al., 2020; Geifman and El-Yaniv, 2017; Guo et al., 2017; Kadavath et al., 2022; Kuhn et al., 2023; Ji et al., 2023)

Table 3 reports 95% bootstrap intervals for the same rows. We use these intervals to separate coarse patterns from small rank differences: the paper-facing claim is that next-action calibration remains unsaturated and boundary errors persist, not that every adjacent model difference is statistically separated.

Figure 3 visualizes the same pattern across overall accuracy, per-slice calibration, and defective-premise over-answering. The key visual result is that the 7B reasoning checkpoint remains weak on false-premise interruption while also abstaining heavily on answerable controls. This is precisely the failure that pooled answer performance would obscure: the assistant can be too reluctant to answer clean questions and still too willing to answer through defective premises.

The prompt-policy controls in Table 2 also separate model capability from prompting effects. The self-contained and compact variants shift error profiles without producing robust gains in utility-aware first-move calibration, matching prior evidence that reasoning-style prompting and iterative prompting can improve some tasks while leaving failure-boundary decisions unstable(Wei et al., 2022; Wang et al., 2023; Yao et al., 2023; Madaan et al., 2023).

The 7B reasoning checkpoint closes the scale/reasoning gate but does not reverse the Day-1 pattern. On the development split, DeepSeek-R1-Distill-Qwen-7B predicts `abstain` for 69/120 examples, `answer` for 46/120, and `challenge` for only 5/120. Its false-premise action accuracy is 0.1, compared with 0.975 for Qwen2.5-1.5B-Instruct

Model	Family	Style	Utility	Action Acc.	Answer Contains	Over-Answer	Challenge Prec.	JSON Parse
Qwen2.5-0.5B-Instruct	Qwen2.5	instruct	-0.4354	0.35	0.05	0.075	0	0.0667
Qwen2.5-1.5B-Instruct	Qwen2.5	instruct	**-0.2229**	**0.7667**	**0.475**	**0.0083**	0.5909	0.8083
DeepSeek-R1-Distill-Qwen-1.5B	DeepSeek/Qwen	reasoning	-0.5125	0.3833	0.15	0.15	0	0.0083
Qwen2.5-Coder-7B-Instruct	Qwen2.5-Coder	instruct	-0.2792	0.6	**0.475**	**0.0083**	**0.9474**	**0.9417**
DeepSeek-R1-Distill-Qwen-7B	DeepSeek/Qwen	reasoning	-0.4313	0.3667	0.3125	0.05	0.8	0.0583

Table 2: Day-1 development comparison for the completed local model matrix. The main decision metrics are utility, action accuracy, and over-answer rate; answer containment and JSON parse rate diagnose content and protocol adherence. Higher is better except over-answer rate.

Model	Utility (95% CI)	Action Acc. (95% CI)	Over-Answer (95% CI)	n
Qwen2.5-0.5B-Instruct	-0.4354 [-0.4729, -0.402]	0.35 [0.2667, 0.4333]	0.075 [0.0333, 0.125]	120
Qwen2.5-1.5B-Instruct	-0.2229 [-0.3167, -0.1271]	0.7667 [0.6917, 0.8417]	0.0083 [0, 0.025]	120
DeepSeek-R1-Distill-Qwen-1.5B	-0.5125 [-0.5583, -0.4624]	0.3833 [0.2998, 0.4669]	0.15 [0.0917, 0.2167]	120
Qwen2.5-Coder-7B-Instruct	-0.2792 [-0.3458, -0.2062]	0.6 [0.5083, 0.6833]	0.0083 [0, 0.025]	120
DeepSeek-R1-Distill-Qwen-7B	-0.4313 [-0.4729, -0.3896]	0.3667 [0.2833, 0.4583]	0.05 [0.0167, 0.0917]	120

Table 3: Bootstrap 95% percentile intervals for the current day-1 scale/reasoning comparison.

and 0.45 for Qwen2.5-Coder-7B-Instruct. At the same time, its answerable-control accuracy is only 0.2, indicating that the model is not merely safer on defective inputs; it also fails to answer many clean supported questions.

Figure 5 makes this boundary error concrete at the action-policy level rather than only in scalar averages. Relative to Qwen2.5-1.5B, the 7B reasoning checkpoint increases abstain-heavy behavior on answerable controls while still under-performing on false-premise correction, which is exactly the coupled calibration failure this benchmark is designed to expose.

The quick+stale split exposes the stale-premise failure more directly. DeepSeek-R1-Distill-Qwen-7B reaches 0.45 action accuracy and -0.475 utility, below Qwen2.5-1.5B-Instruct and Qwen2.5-Coder-7B-Instruct. On the stale-premise slice, the 7B reasoning checkpoint has 0.25 action accuracy and 0.75 over-answer rate, meaning that it often answers through an outdated premise instead of challenging it.

External API contrast. To test whether the main pattern is an artifact of local checkpoints, we ran five external API baselines on the same 560-example canonical split with the same prompt protocol (`decision_first`, max_tokens=64). Table 5, Table 6, and Figure 6 show a non-trivial frontier spread: `gpt-5-chat-latest` reaches the best action accuracy (0.9286), `qwen-plus-latest` reaches the best utility (0.1571), and `qwen-turbo` remains strong at lower cost. We read these API rows as uncertainty-qualified point estimates rather than as a fine-grained significance ranking. The main claim is the persistent boundary difficulty: `gpt-5-chat-latest` still over-answers 29.17% of false-premise cases even when overall action accuracy is high.

Slice-balance sensitivity. The canonical split keeps fixed action totals but leaves slice gaps (`answerable_control` and `conflicting_evidence`). To test whether this imbalance drives the API conclusion, we resumed only the 40 additional rows in a slice-balanced 600-example split for a frontier API, two stronger APIs, and one lower-cost API. Table 7 reports full-minus-canonical deltas under matched decoding, with bootstrap intervals for action accuracy and utility. For `qwen-plus-latest`, both `decision_first` and `decision_first_guarded` have positive action-accuracy point deltas (+0.0096 and +0.0079), but only `decision_first` improves utility (+0.0212), while guarded lowers utility (-0.0372). `gpt-5-chat-latest`, `gpt-4.1-mini`, and `qwen-turbo` show the same action-accuracy point-estimate direction (+0.0047, +0.0076, and +0.0107) while utility decreases (-0.0363, -0.0371, and -0.0357). Because the delta intervals overlap zero, we use the 600-example split as sensitivity evidence rather than as a replacement benchmark: the point estimates do not suggest a reversal, but the headline remains the canonical 560-example split.

7 Decision-First Intervention

This intervention is deliberately lightweight: it modifies instruction-level decision ordering rather

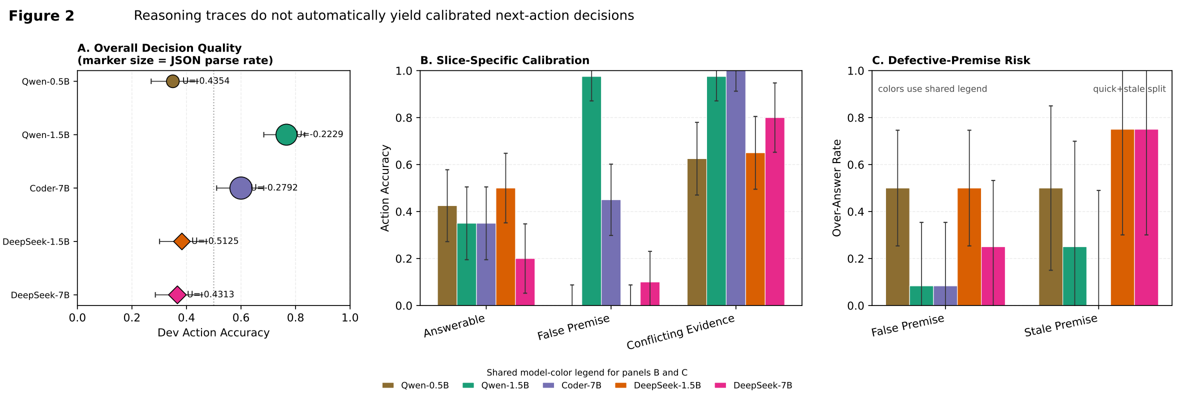

Figure 3: Reasoning traces do not automatically yield calibrated next-action decisions. Panel A shows development action accuracy for the completed model matrix, with marker size proportional to JSON parse rate and labels giving average utility. Panel B reports per-slice action accuracy for answerable, false-premise, and conflicting-evidence slices to show where scale helps and where boundary control fails. Panel C shows over-answer rates on false- and stale-premise items in quick+stale with a shared model-color legend below the panels. The shared pattern is that caution is not monotonic safety: the 7B reasoning checkpoint is weak on false/stale premise interruption while also abstaining heavily on clean answerable controls.

Model	Answerable	False Premise	Conflicting Evidence
Qwen2.5-0.5B-Instruct	-0.4188 / 0.425	-0.4188 / 0	-0.4688 / 0.625
Qwen2.5-1.5B-Instruct	-0.625 / 0.35	0.4625 / 0.975	-0.5062 / 0.975
DeepSeek-R1-Distill-Qwen-1.5B	-0.4688 / 0.5	-0.5875 / 0	-0.4813 / 0.65
Qwen2.5-Coder-7B-Instruct	-0.4062 / 0.35	0.0688 / 0.45	-0.5 / 1
DeepSeek-R1-Distill-Qwen-7B	-0.5062 / 0.2	-0.2875 / 0.1	-0.5 / 0.8

Table 4: Per-slice utility / action-accuracy pairs on Day-1 development. The slice table shows that model errors are structured by input defect type rather than explained by a single pooled score.

than adding a new training stage or external verifier, following prompt-level calibration ideas from prior reasoning and self-consistency work(Wei et al., 2022; Wang et al., 2023; Madaan et al., 2023).

The intervention results suggest that action-selection failures are not fixed simply by adding more critique language. On Qwen2.5-1.5B quick+stale, `decision_first` improves utility from -0.2188 to -0.1375, action accuracy from 0.775 to 0.85, and over-answer rate from 0.05 to 0. Crucially, this gain does not come from sacrificing answer-supported behavior: answer-supported accuracy changes from 0.7083 to 0.75, while defective-premise accuracy rises from 0.875 to 1. In contrast, `critique_first`, `decision_first_guarded`, and `decision_first_balanced` either over-challenge answerable items or collapse answer-supported accuracy. This supports a narrow method claim: making the action decision explicit can help, but longer or more cautious prompts are not automatically better.

8 Qualitative Analysis

Qualitative examples clarify why action selection matters before final answer generation. Table 9 shows four boundary cases from the active evidence bundle. In a stale-premise example, the user asks why Facebook is still trading under `FB`; the correct next action is `challenge`, because Meta's stock began trading under `META` in 2022. Qwen2.5-1.5B instead answers while mentioning the correction. The utility loss is not lack of factual knowledge; it is accepting the user's outdated framing as an answer request (Kasai et al., 2023; Vu et al., 2023).

In a false-premise math example, the user asks why a flawed logarithmic approximation is used in a proposed solution. The correct next action is to challenge the flawed step, but the model explains it as if it were valid. This is the central action-boundary failure for reasoning-style responses: fluent intermediate reasoning can rationalize a bad premise instead of interrupting it (Lin et al., 2022; Ji et al., 2023).

The 7B reasoning checkpoint shows the com-

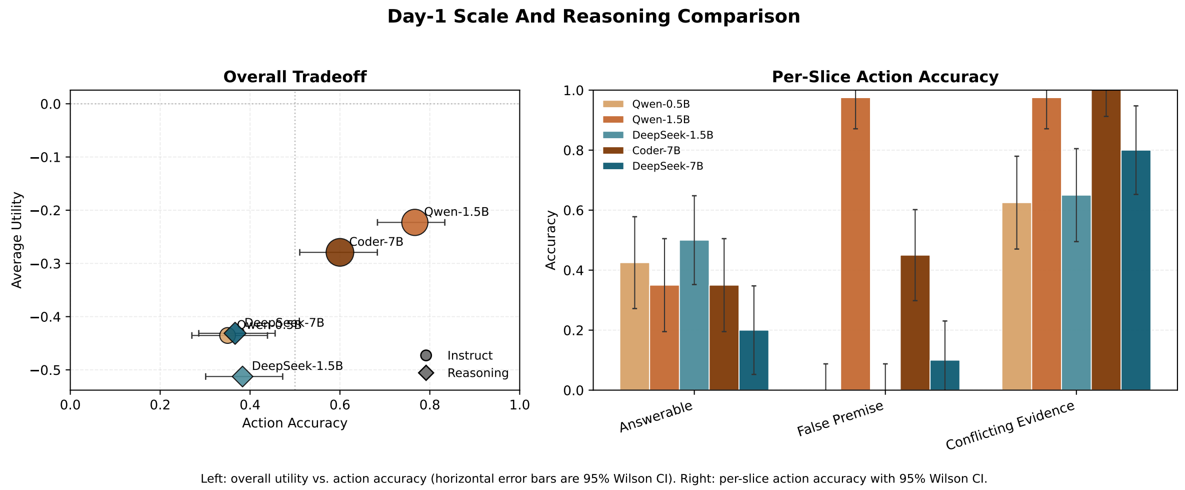

Figure 4: Scale-reasoning tradeoff on the Day-1 development split. The left panel plots action accuracy against utility, with marker size indicating JSON parse rate. The right panel reports per-slice action accuracy across answerable, false-premise, and conflicting-evidence slices to show where scale helps and where boundary control fails.

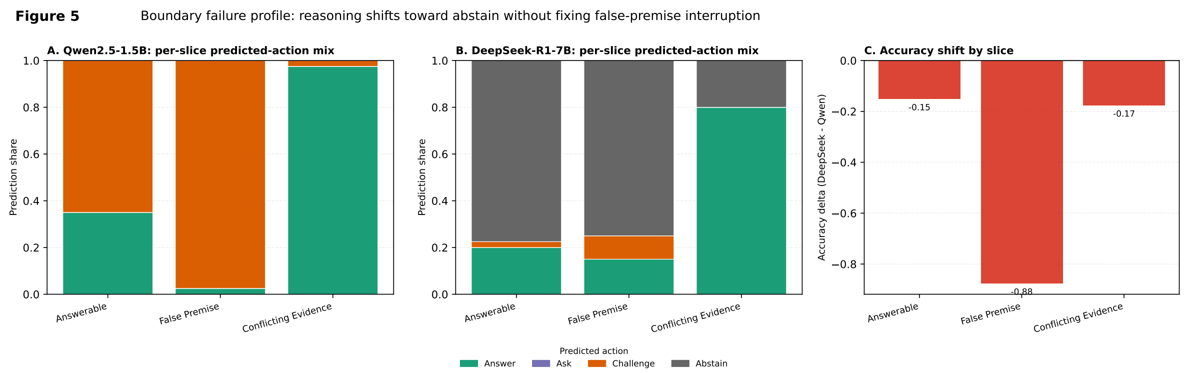

Figure 5: Boundary failure profile on the Day-1 development split. Panels A and B show per-slice predicted-action distributions for Qwen2.5-1.5B-Instruct and DeepSeek-R1-Distill-Qwen-7B. Panel C shows per-slice action-accuracy deltas (DeepSeek minus Qwen). The key pattern is asymmetric: reasoning shifts toward abstain on answerable controls without recovering false-premise interruption, so caution does not translate into better first-move calibration.

plementary risk. On a clean answerable triangulation problem, it produces extended reasoning but predicts `abstain` rather than committing to the supported answer. On a false-premise tournament problem, it reasons about the stated premise but does not explicitly correct the false 3^t construction to the supported 2^t construction. These cases demonstrate that the benchmark does not reward caution or reasoning length alone; the model must choose the right repair action for the input state (Kamath et al., 2020; Kadavath et al., 2022). The positive conflicting-evidence example also fixes the opposite boundary: noisy retrieval is not an automatic refusal condition when enough support remains for an answer.

9 Limitations

This submission reports an initial curated benchmark rather than a final large-scale evaluation. The canonical split now includes all four actions and six slices, including ambiguous-intent and insufficient-evidence cases. The strongest empirical evidence is still concentrated on answer-vs-challenge boundaries, stale-premise correction, and retrieval-conflict calibration, where source diversity and model-analysis depth are most mature. Claims about `ask` and hard `abstain` should therefore remain slice-scoped rather than presented as fully balanced evidence across all possible clarification and abstention settings. The utility metric is intended as a diagnostic summary and is always re-

Model	Input ($/M)	Output ($/M)	Action Acc.	Avg Utility	JSON Parse	Over-answer	Answer Contains
qwen-turbo	$0.046	$0.092	0.8393	0.0353	0.7625	0.0500	0.6550
gpt-4o-mini	$0.1500	$0.6000	0.7679	0.0799	0.8946	0.0089	0.4500
gpt-4.1-mini	$0.1500	$0.6000	0.8857	0.0558	0.8500	0.0411	0.6700
qwen-plus-latest	$0.1100	$0.2750	0.8571	0.1571	0.8054	0.0304	0.6200
gpt-5-chat-latest	$1.2500	$10.0000	0.9286	0.0442	0.8554	0.0625	0.7650

Table 5: External API baselines on the 560-example canonical split (decision-first, max_tokens=64, temperature=0.0). Metrics are measured on `avg_utility`, `action_accuracy`, `json_parse_rate`, `over_answer_rate`, and `answer_contains_rate`.

Model	Utility (95% CI)	Action Acc. (95% CI)	Over-Answer (95% CI)	n
qwen-turbo	0.0353 [-0.0112, 0.0786]	0.8393 [0.8107, 0.8696]	0.05 [0.0339, 0.0679]	560
gpt-4o-mini	0.0799 [0.0357, 0.121]	0.7679 [0.7321, 0.8018]	0.0089 [0.0018, 0.0179]	560
gpt-4.1-mini	0.0558 [0.0147, 0.0973]	0.8857 [0.8589, 0.9107]	0.0411 [0.0268, 0.0589]	560
qwen-plus-latest	0.1571 [0.1134, 0.2031]	0.8571 [0.8268, 0.8857]	0.0304 [0.0179, 0.0464]	560
gpt-5-chat-latest	0.0442 [-0.0027, 0.0875]	0.9286 [0.9071, 0.9483]	0.0625 [0.0446, 0.0821]	560

Table 6: Bootstrap 95% percentile intervals for external API baselines on the canonical split.

Setting	Δ Action Acc. (95% CI)	Δ Utility (95% CI)	Δ Over-answer	Δ JSON Parse
qwen-plus-latest / decision-first	0.0096 [-0.0314, 0.0476]	0.0212 [-0.0415, 0.0844]	-0.0021	0.0129
qwen-plus-latest / guarded	0.0079 [-0.0264, 0.0432]	-0.0372 [-0.0933, 0.0213]	-0.0024	-0.0036
gpt-4.1-mini / decision-first	0.0076 [-0.0282, 0.0457]	-0.0371 [-0.0989, 0.0193]	-0.0028	0.0100
qwen-turbo / decision-first	0.0107 [-0.0321, 0.0524]	-0.0357 [-0.0978, 0.0263]	-0.0033	0.0158
gpt-5-chat-latest / decision-first	0.0047 [-0.0232, 0.0341]	-0.0363 [-0.0976, 0.0227]	-0.0042	0.0096

Table 7: Stress sensitivity from canonical 560 to slice-balanced 600, not a replacement headline benchmark, under fixed decoding settings (`max_tokens`=64, temperature=0). Values are Δ (full minus canonical); action-accuracy and utility columns include bootstrap 95% percentile intervals.

ported alongside action accuracy, over-answer rate, per-slice metrics, and confusion patterns; claims should rest on agreement between the summary score and these disaggregated behaviors. We also use saved-output utility-weight sensitivity audits as claim guardrails rather than as replacement metrics.

The slice-balanced 600-example variant is a stress split, not the canonical benchmark. It adds answerable-control and conflicting-evidence rows to probe the residual slice imbalance in the 560-example split, but it also changes the answer count and is therefore used only for sensitivity analysis. Headline benchmark claims remain tied to the canonical 560-example split unless a future version re-validates and freezes a new canonical composition.

The bootstrap intervals are local uncertainty checks over the current split samples. They do not capture uncertainty from example construction, prompt-policy choice, or the decision to treat the 600-example variant as stress evidence. We therefore use them to guard against over-reading point estimates rather than as proof of a stable model ordering.

The DeepSeek reasoning runs also show low JSON adherence. We report this as part of the observed assistant behavior under the protocol, because the task requires an explicit action decision, but saved-output parse-sensitivity audits show that low-adherence rows are protocol-sensitive. As a result, broader claims about reasoning models should remain scoped to this prompt and parsing setup until additional format-control runs are complete.

This draft also focuses on action choice before response generation, not on external tool-use policy. Tool-augmented behaviors can change error recovery patterns in practice, but they are not benchmarked in the current split and should be treated as future expansion work rather than implied evidence in this paper(Schick et al., 2023).

Human validation supports the active labels and submission arithmetic under the documented ontology. It does not prove that the ontology is the only possible formulation of assistant action choice. Borderline examples and alternate taxonomies remain important future work.

Responsible deployment also requires care. A system that overuses `challenge` or `abstain` can become obstructive, especially for users asking ordinary answerable questions, using nonstandard phrasing, writing in a second language, or expressing low confidence. Conversely, a system that overuses `answer` can reinforce false or stale assumptions. We therefore treat the benchmark as an evaluation of action calibration, not as a prescrip-

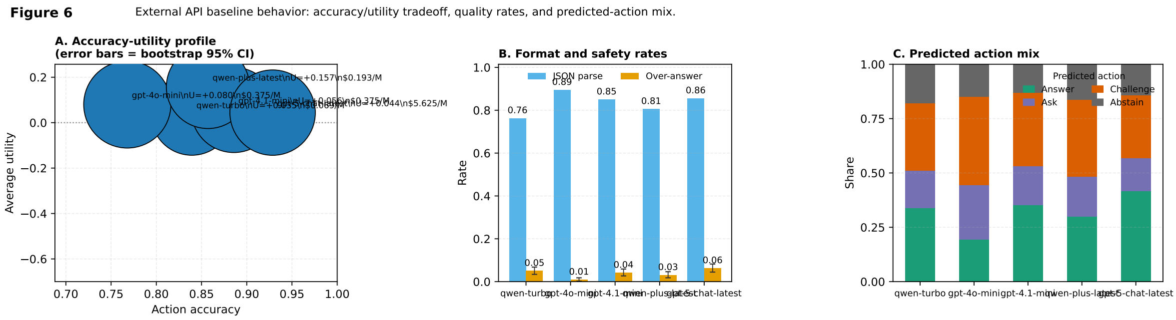

Figure 6: API baseline comparison on the 560-example canonical split with decision-first prompting. The left panel plots utility against action accuracy, where marker size indicates JSON parse rate and error bars show bootstrap 95% intervals; the middle panel compares format/safety rates with bootstrap intervals for over-answer rate; and the right panel shows the predicted action mix. The pattern aligns with Tables 5–6: stronger APIs extend the observed accuracy–utility tradeoff, but boundary slices remain the main error source.

tion that assistants should be more confrontational in all settings.

The released artifacts should avoid exposing private user data: all current examples are converted from benchmark or curated public-source material. Before submission, the anonymized package should still be checked for local paths, model-cache paths, and process notes that are useful internally but inappropriate for a double-blind review package.

References

Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. 2019. Asking clarifying questions in open-domain information-seeking conversations. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 475–484, Paris, France. Association for Computing Machinery.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, pages 4885–4894.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. 2020. Retrieval augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning*.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. *International Conference on Learning Representations*.

Gautier Izacard and Edouard Grave. 2021. Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, and

Variant	Utility	Δ Utility	Action Acc.	Δ Acc.	Over-Answer	Answer-Supported Acc.	Defective-Premise Acc.
baseline	-0.2188	0	0.775	0	0.05	0.7083	0.875
decision_first	**-0.1375**	+0.0813	**0.85**	+0.075	**0**	0.75	**1**
critique_first	-0.2812	-0.0624	0.4	-0.375	0.025	0.0416	0.9375
decision_first_guarded	-0.1812	+0.0376	0.625	-0.15	**0**	0.5	0.8125
decision_first_balanced	-0.1313	+0.0875	0.375	-0.4	**0**	0	0.9375

Table 8: Prompt-intervention pilot on the Qwen2.5-1.5B quick+stale split. Deltas are relative to the baseline prompt. Answer-supported accuracy pools answerable controls and conflicting-evidence items; defective-premise accuracy pools false- and stale-premise items. Utility is highlighted only among variants that do not reduce overall or answer-supported action accuracy versus baseline.

Case	Prompt signal	Gold action	Model action	Action-selection lesson
Stale premise	User asks why Facebook is still trading under `FB`; evidence says Meta changed the ticker to `META` on June 9, 2022.	`challenge`	`answer`	The model mentions the correction but still answers through the stale premise. Factual knowledge is present, but the next action is wrong.
False premise	A math-solution step uses $1 + 2/(3n+1)$ where the supported step is $1 + 1/(3n + 1)$.	`challenge`	`answer`	The model explains why logarithms are useful instead of interrupting the defective step. Fluent explanation can rationalize a bad premise.
Answerable control	Jenny has 36 cookies; her son eats 20 and her husband eats 7.	`answer`	`challenge`	The supported answer is 9, but the model claims the prompt is incomplete. The task does not reward caution when the input is already answerable.
Conflict but answerable	Noisy passages ask who was told to beware the Ides of March. Several passages still support Caesar.	`answer`	`answer`	Retrieval noise alone is not an abstention condition. The action depends on whether enough support remains.

Table 9: Representative action-boundary examples from the Day-1 evidence bundle. The table reports the next-action label rather than grading only the final answer text.

17 others. 2022. Language models (mostly) know what they know. *Preprint*, arXiv:2207.05221.

Amita Kamath, Robin Jia, and Percy Liang. 2020. Selective question answering under domain shift. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 5684–5696, Online. Association for Computational Linguistics.

Jungo Kasai, Keisuke Sakaguchi, Yoichi Takahashi, Ronan Le Bras, Akari Asai, Xinyan Velocity Yu, Dragomir Radev, Noah A. Smith, Yejin Choi, and Kentaro Inui. 2023. RealTime QA: What's the answer right now? In *Advances in Neural Information Processing Systems*.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations*.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452–466.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, and 1 others. 2023. Holistic evaluation of language models. *Transactions on Machine Learning Research*.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.

Siyi Liu, Qiang Ning, Kishaloy Halder, Zheng Qi, Wei Xiao, Phu Mon Htut, Yi Zhang, Neha Anna John, Bonan Min, Yassine Benajiba, and Dan Roth. 2025. Open domain question answering with conflicting contexts. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1838–1854, Albuquerque, New Mexico. Association for Computational Linguistics.

Aman Madaan, Niket Tandon, Peter Clark, Yiming Yang, Shyam Upadhyay, Jena Lee, Caiming Gao, Yiming Gu, and Sameer Singh. 2023. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*.

Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Jacob Menick, Maja Trebacz, Vladimir Mikulik, John Aslanides, Francis Song, Martin Chadwick, Mia Glaese, Susannah Young, Lucy Campbell-Gillingham, Geoffrey Irving, and Nat McAleese. 2022. Teaching language models to support answers with verified quotes. *Preprint*, arXiv:2203.11147.

Sewon Min, Julian Michael, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2020. AmbigQA: Answering ambiguous open-domain questions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5783–5797, Online. Association for Computational Linguistics.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. 2021. WebGPT: Browser-assisted question-answering with human feedback. *Preprint*, arXiv:2112.09332.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*.

Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know what you don't know: Unanswerable questions for SQuAD. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 784–789, Melbourne, Australia. Association for Computational Linguistics.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*.

Sagi Shaier, Ari Kobren, and Philip V. Ogren. 2024. Adaptive question answering: Enhancing language model proficiency for addressing knowledge conflicts with source citations. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17226–17239, Miami, Florida, USA. Association for Computational Linguistics.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, and 1 others. 2023. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*.

Ivan Stelmakh, Yi Luan, Bhuwan Dhingra, and Ming-Wei Chang. 2022. ASQA: Factoid questions meet long-form answers. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 8273–8288, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: A large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, New Orleans, Louisiana. Association for Computational Linguistics.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong. 2023. FreshLLMs: Refreshing large language models with search engine augmentation. *Preprint*, arXiv:2310.03214.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*.

Xinyan Yu, Sewon Min, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. CREPE: Open-domain question answering with false presuppositions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10457–10480, Toronto, Canada. Association for Computational Linguistics.

Michael J. Q. Zhang and Eunsol Choi. 2021. SituatedQA: Incorporating extra-linguistic contexts into QA. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7371–7387, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.